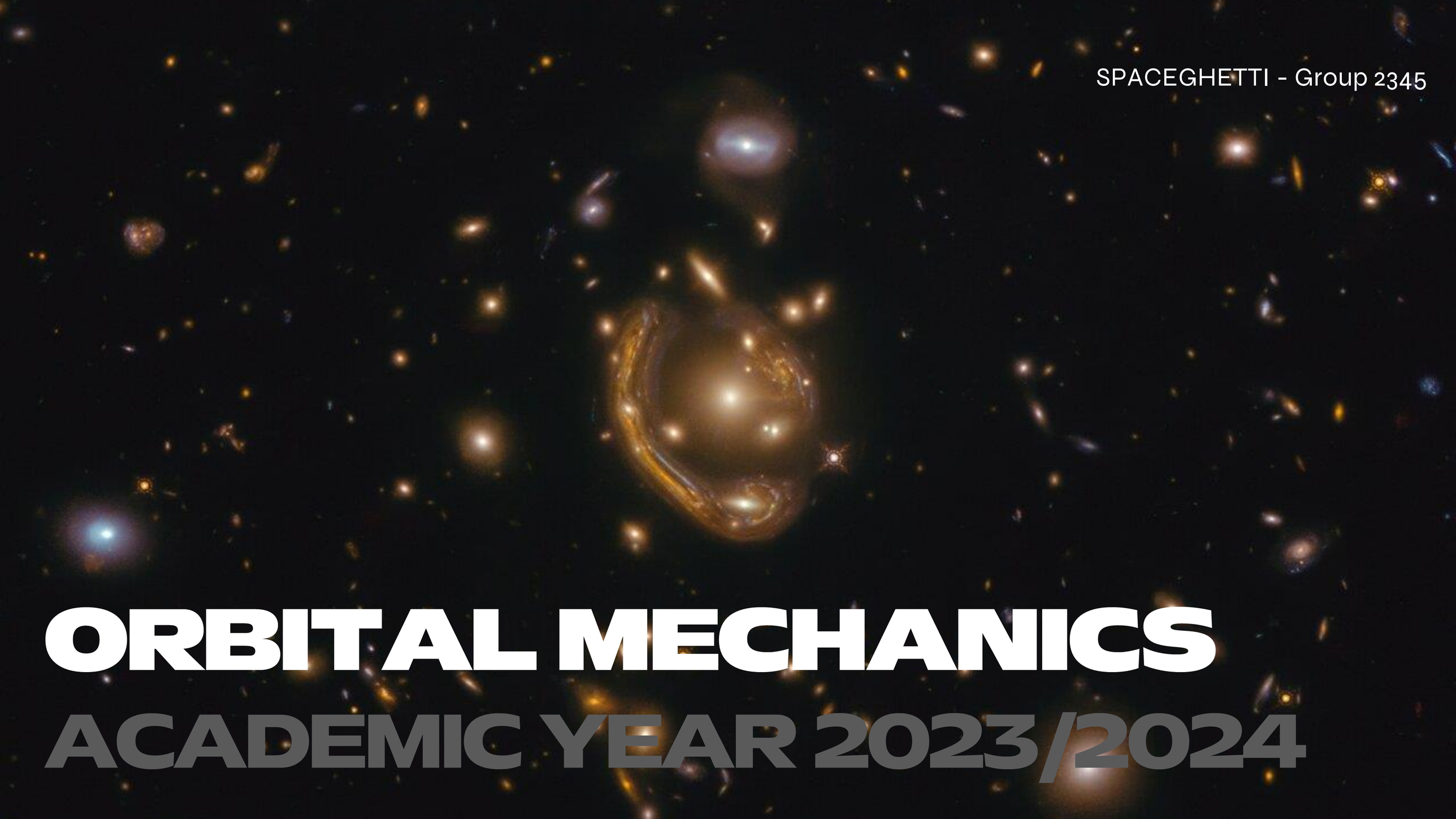


SPACEGHETTI - Group 2345

ORBITAL MECHANICS

ACADEMIC YEAR 2023/2024

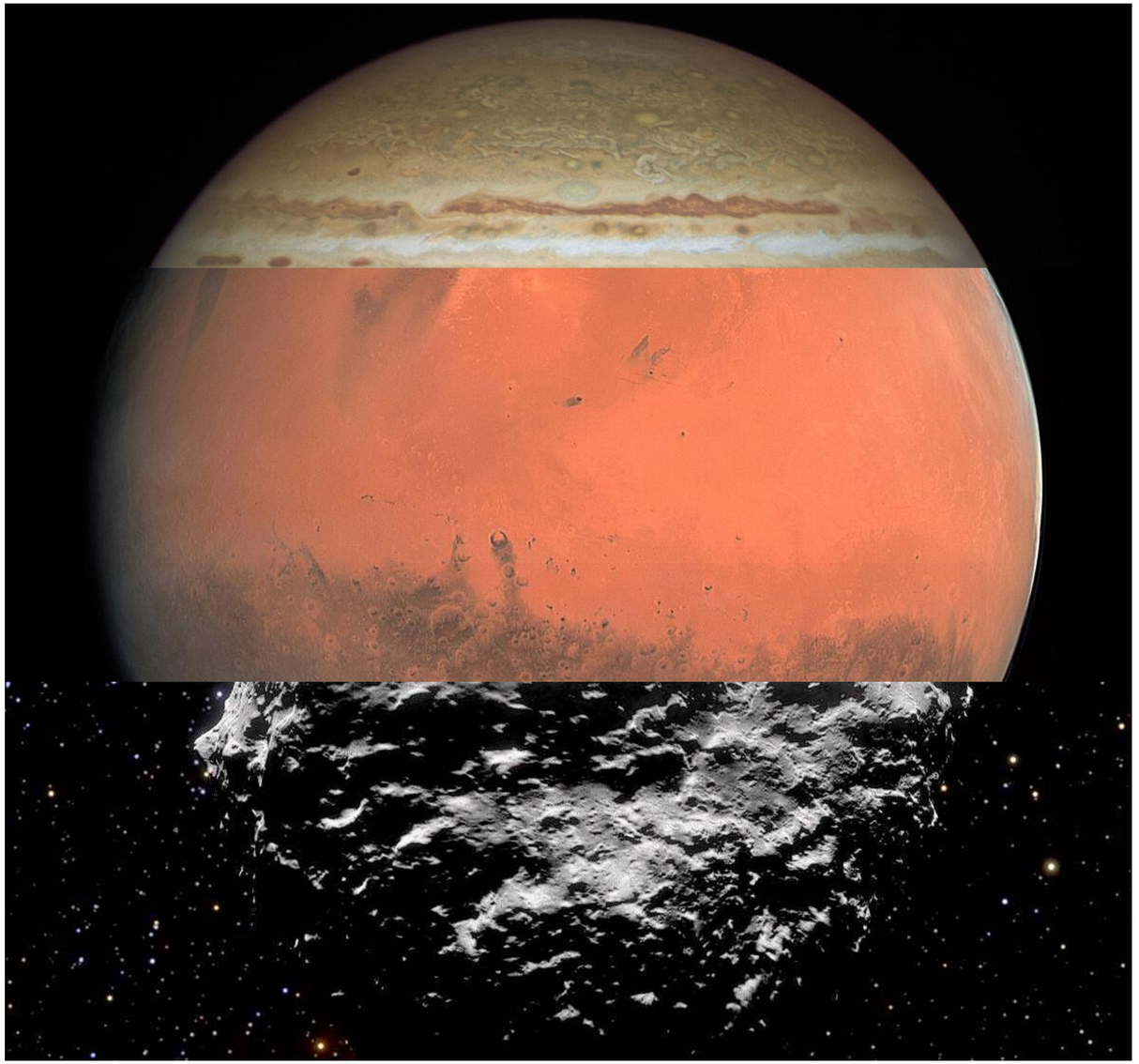
INTERPLANETARY EXPLORER MISSION

INTRODUCTION

The mission objective is performing an interplanetary transfer:

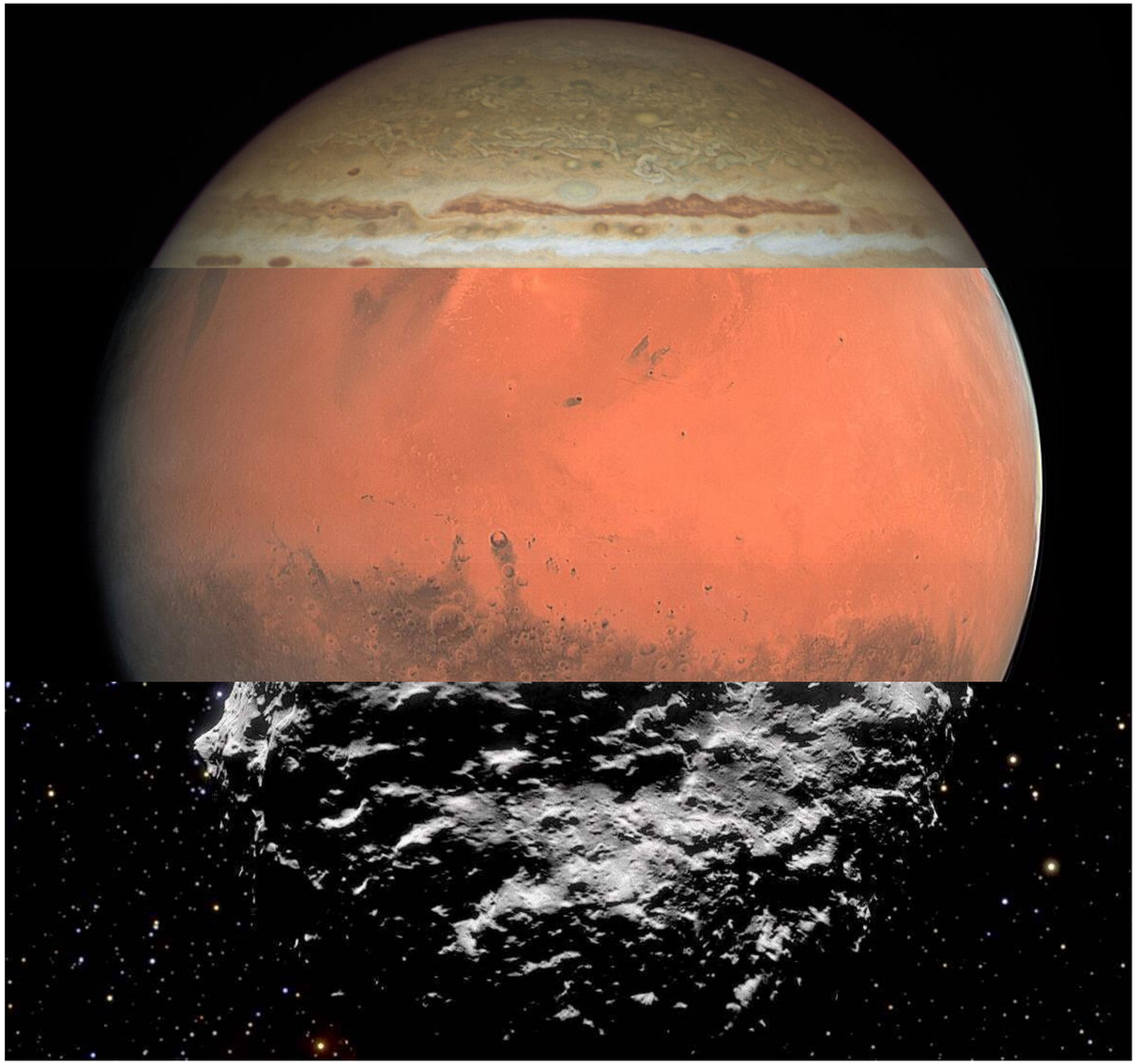
- departure planet: Jupiter
- intermediate powered gravity assist at: Mars
- arrival NEO: 1996FG3 [2]

The objective of the mission design is the minimizzation fo the total Δv.



CONSTRAINTS

- earliest departure date: 2028/01/01 00:00:00;
- latest arrival date: 2058/01/01 00:00:00;
- minimum radius of pericenter for the powered gravity assist around Mars: 250 km [3].



ASSUMPTIONS

Patched conics method

- SOI of each planet in their planetocentric frame is supposed infinite;
- SOI of each planet in the heliocentric frame is supposed negligible
- flyby time is considered negligible, with a supposed uniform gravitational field around the planet;
- initial heliocentric orbit is equal to that of the departure planet;
- final heliocentric orbit is equal to that of the arrival asteroid;
- all maneuvers are assumed to be instantaneous; Possible encounters with other planets and Solar Radiation Perturbation are ignored.

TIME WINDOWS ANALYSIS

Orbital periods and synodic periods

To define the time windows a consideration on the orbital periods and synodic periods has been made.

$T_J[days]$	$T_M[days]$	$T_A[days]$	$T_{Syn,JM}[days]$	$T_{Syn,MA}[days]$	$T_{Syn,JA}[days]$
4334.3522	686.9777	395.2386	816.3690	930.6948	434.8956

Homann transfer

The limitation of the flyby and arrival window has been made considering a generalized Homann transfer.

Time windows

	Start	End
Departure time window	2028/1/1 - 0:0:0	2039/11/13 - 8:27:10
FlyBy time window	2030/2/27 - 11:5:42	2043/11/16 - 15:20:36
Arrival time window	2030/7/25 - 20:1:43	2044/8/18 - 4:30:22

GRID SEARCH METHOD

Algorithm

```
for l = 1 : Nloops
    % In this section the time windows were recomputed based
    % on the minimum found in the previous iteration
    for i = 1 : Npoints
        for j = 1 : Npoints
            for k = 1 : Npoints
                Dv(i, j, k) = (departure(i), flyby(j), arrival(k))
            end
        end
    end
    % Get the minimum, to update the time windows
end
```

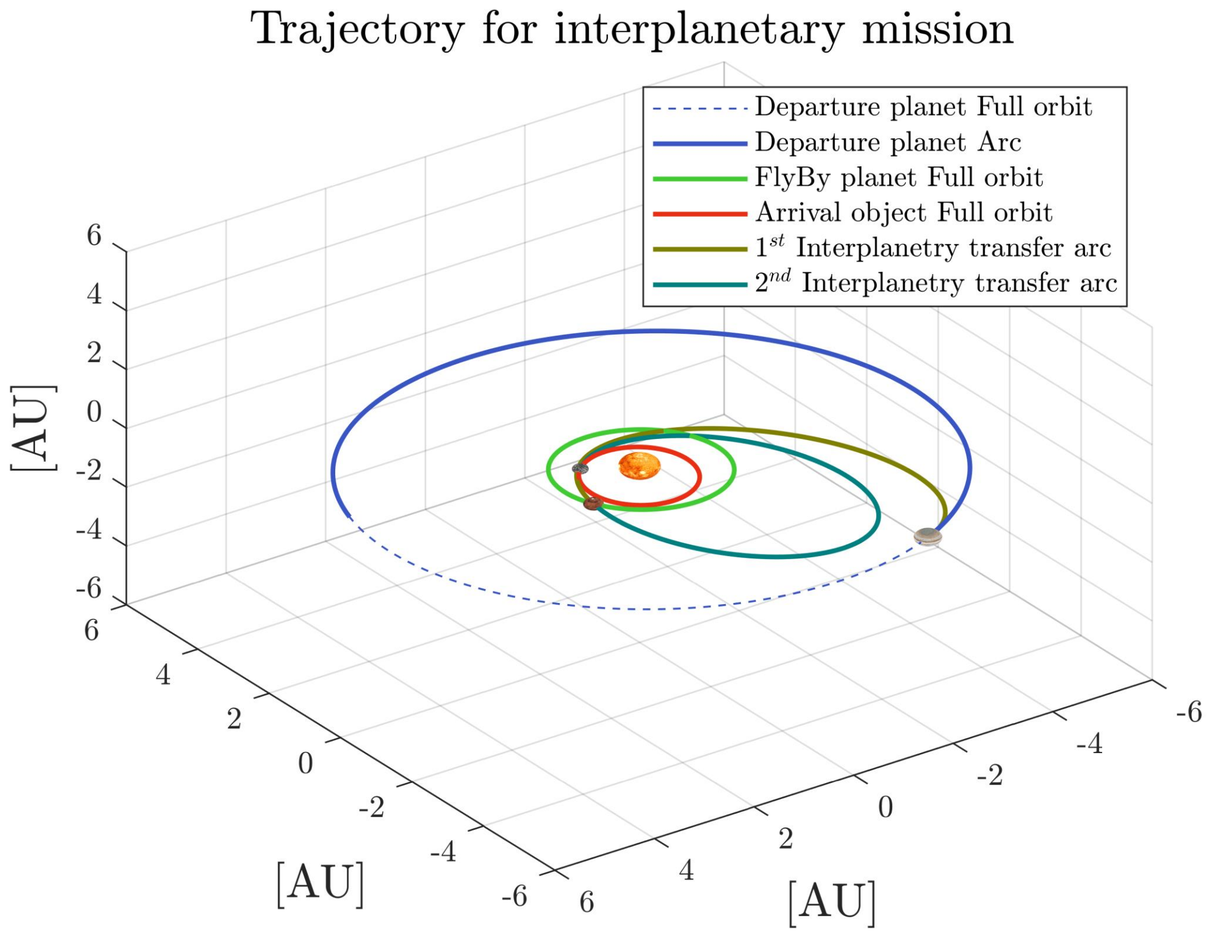
Results

$N_{points}[-]$	t_{dep} [mjd2000]	t_{fb} [mjd2000]	t_{arr} [mjd2000]	$\Delta t[days]$	$\Delta v[km/s]$	$t_{comp}[s]$
50	11457.3707	12653.1121	13885.0403	2427.6696	14.2951	84.220146
100	10398.0875	11878.5611	13812.7221	3414.6347	14.3804	660.429101
200	10714.6166	11895.8242	13429.6118	2714.9953	13.0748	4997.314121

RESULTS

Departure	FlyBy	Arrival	$\Delta v_{dep}[km/s]$	$\Delta v_{ga}[km/s]$	$\Delta v_{arr}[km/s]$
2029/5/3 2:47:51	2032/7/27 7:46:47	2036/10/8 2:41:3	5.5637	1.2067e-08	7.5111

	$a[km]$	$e[-]$	$i[deg]$	$\Omega[deg]$	$\omega[deg]$	$\theta_i[deg]$	$\theta_f[deg]$
1st Interplanetary arc	4.7844e+08	0.7062	2.0747	61.7179	324.0143	178.3709	91.2068
2nd Interplanetary arc	4.0136e+08	0.6794	1.9072	0.2213	17.1691	99.5182	45.0262



FLYBY

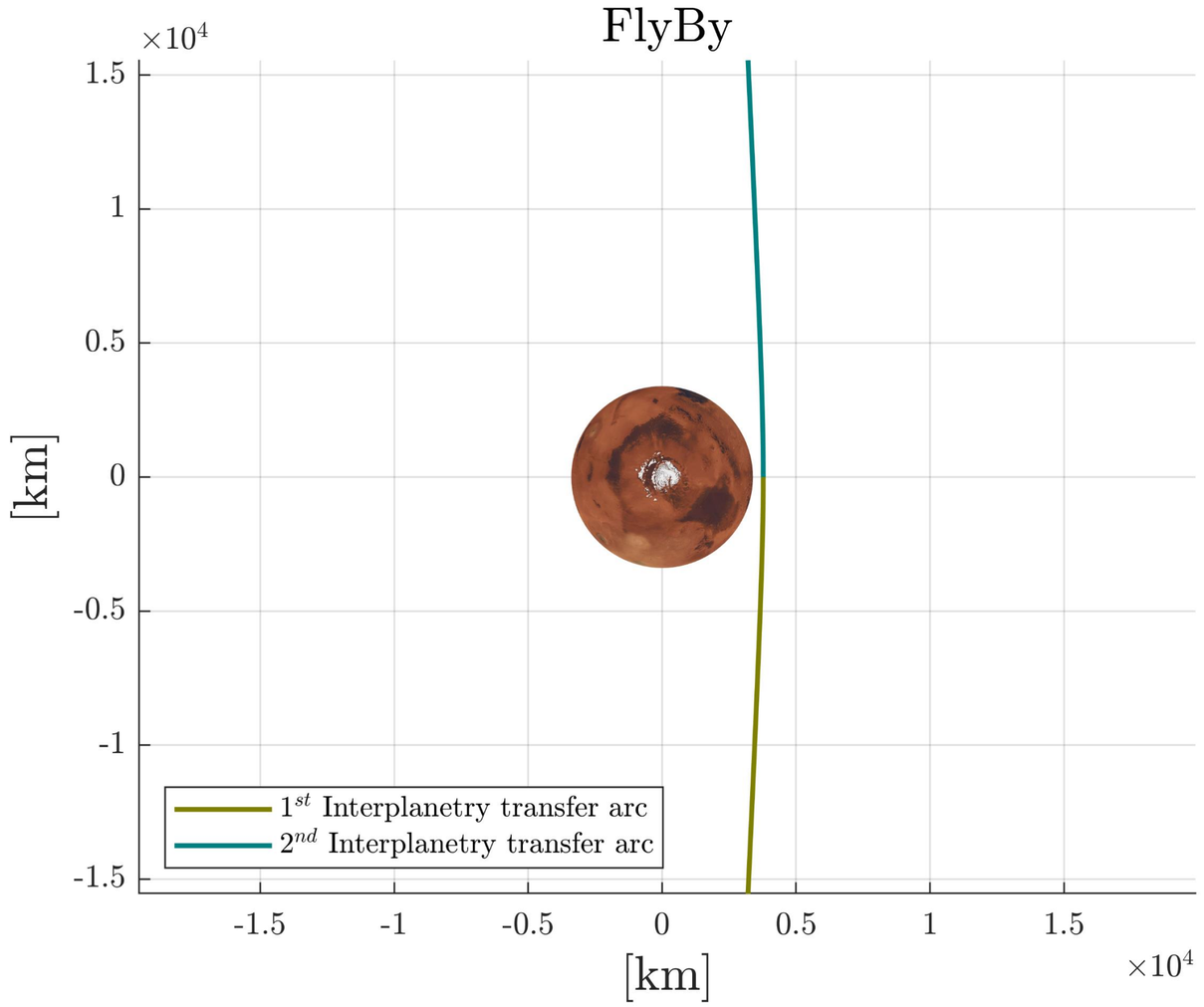
	Incoming hyperbola	Outgoing hyperbola
a $[km]$	-185.9012	-185.9012
e $[-]$	21.3045	21.3045
hp $[km]$	384.7253	384.7253
v_p $[km/s]$	15.9083	15.9083
v_∞ $[km/s]$	15.1783	15.1783

The Δv_{fb} which is given by the difference between the incoming and outgoing heliocentric velocities of the spacecraft is equal to:

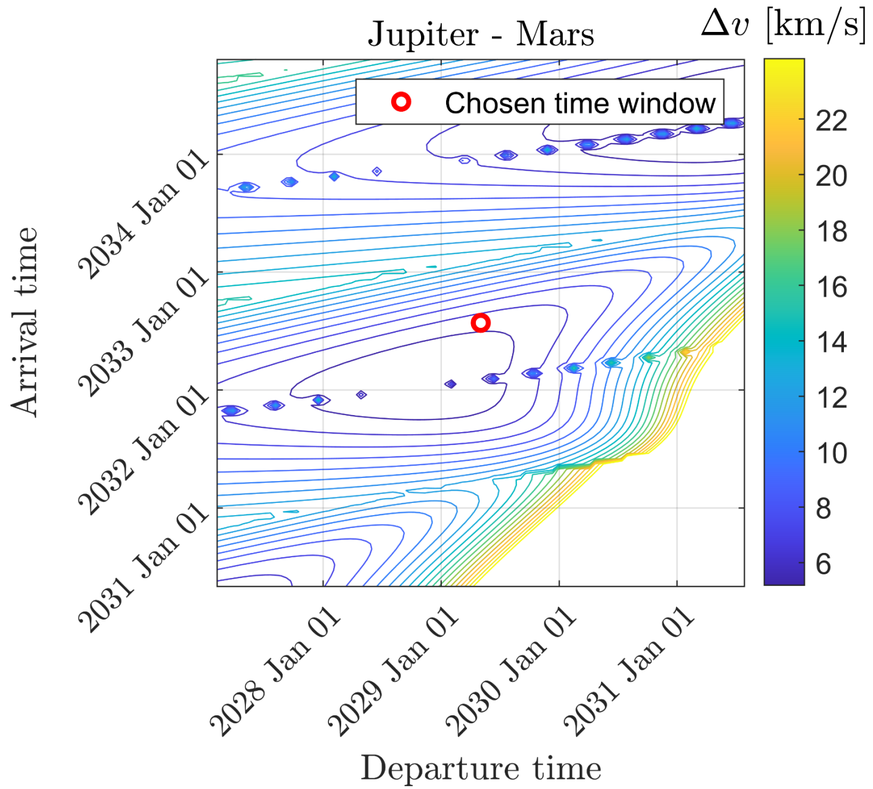
$$\Delta v_{flyby} = 1.4249\, km/s \qquad \Delta v_{ga}/\Delta v_{flyby} = 8.4684e-09$$

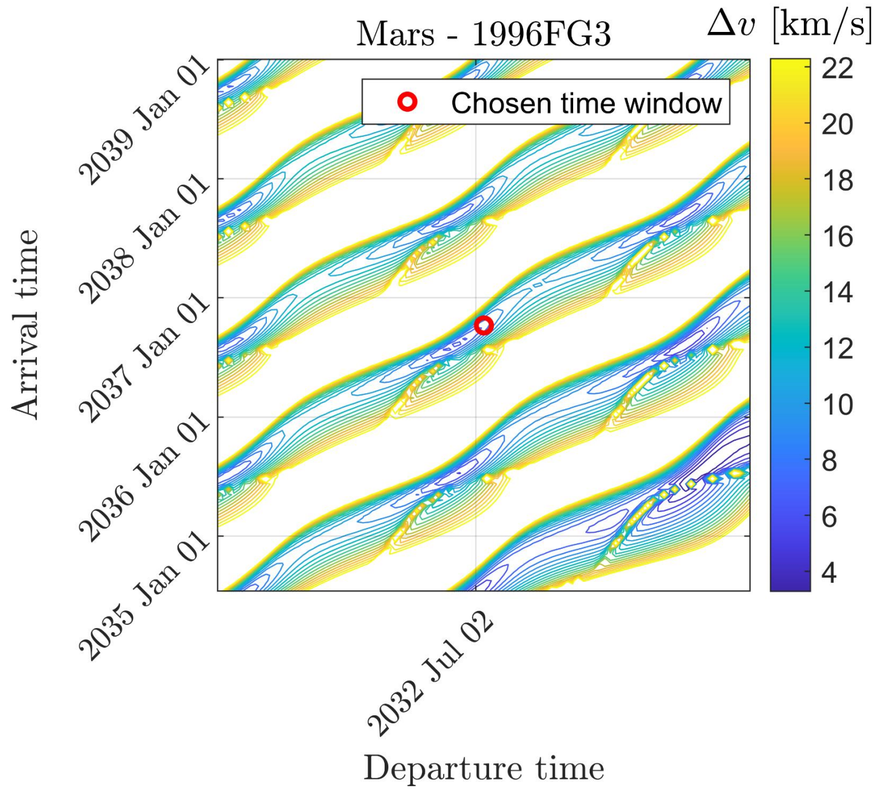
The total duration of the fly-by, considering a finite SOI is:

$$\Delta t_{fb} = 22.5321\, h$$



DISCUSSION





- Most expensvive maneuver: injection in 1996FG3 orbit
- Least expensive maneuver: powered gravity assist

PLANETARY MISSION

INTRODUCTION

The mission objective is to study:

- the ground track for an unperturbed and a perturbed orbit;
- the comparison between cartesian method and Gauss planetary equations;
- the secular effects for the keplerian elements due to the perturbations;
- the comparison between cartesian and NASA Horizions System [6].

$a[km]$	$e[-]$	$i[deg]$	$\Omega[deg]$	$\omega[deg]$	$\theta_i[deg]$	T[h]	k	n
7192	0.0493	82.395	0	0	0	1.6861	14	1



PERTURBATIONS

J2

Earth's oblateness and spheroid structure results in a non-uniform gravitational field. The zonal effect of the real gravitational field can be modeled using a series of spherical harmonics, in which the dominating term is J2.

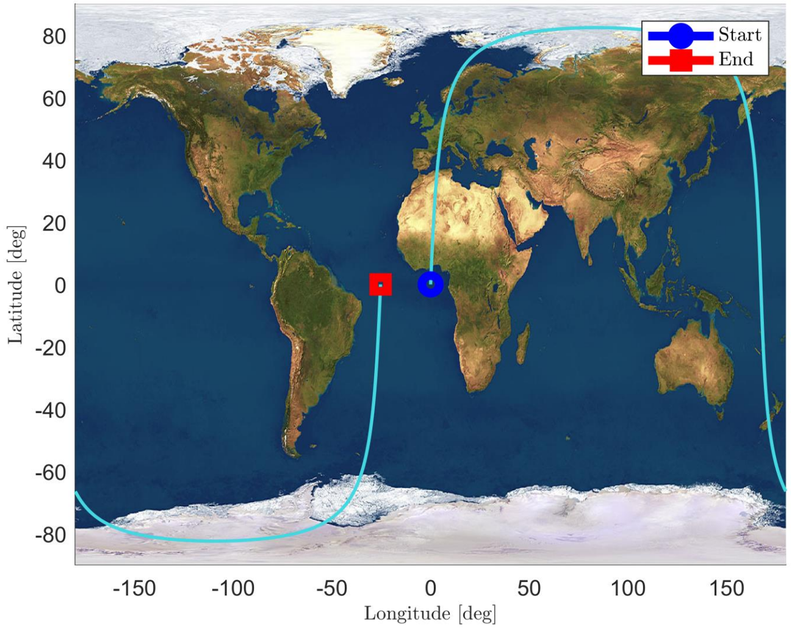
$$\underline{a}_{J2} = \frac{3}{2}\frac{J2\mu_E R_E}{r^4}\left[\frac{x}{r}\left(5\frac{z^5}{r^2}-1\right);\quad \frac{y}{r}\left(5\frac{z^5}{r^2}-1\right);\quad \frac{z}{r}\left(5\frac{z^5}{r^2}-3\right)\right]$$

Aerodynamic drag

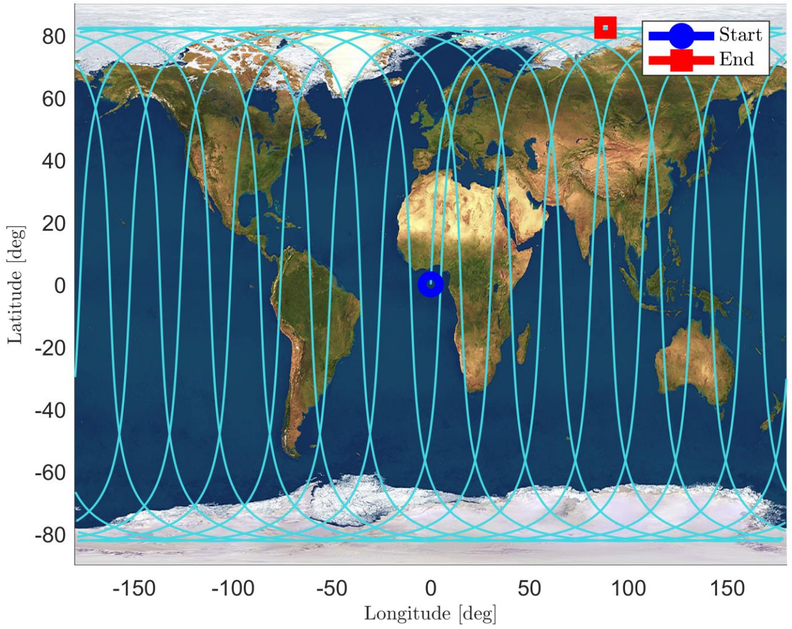
The second major effect, for an orbit characterized by a low altitude, is the aerodynamic drag due to the top layers of the Earth atmosphere.

$$\underline{a}_{Drag} = -\frac{1}{2}\frac{A_{cross}C_D}{m}\rho(h,t)\underline{v}_{rel}{}^2\frac{v_{rel}}{||\underline{v}_{rel}||}$$

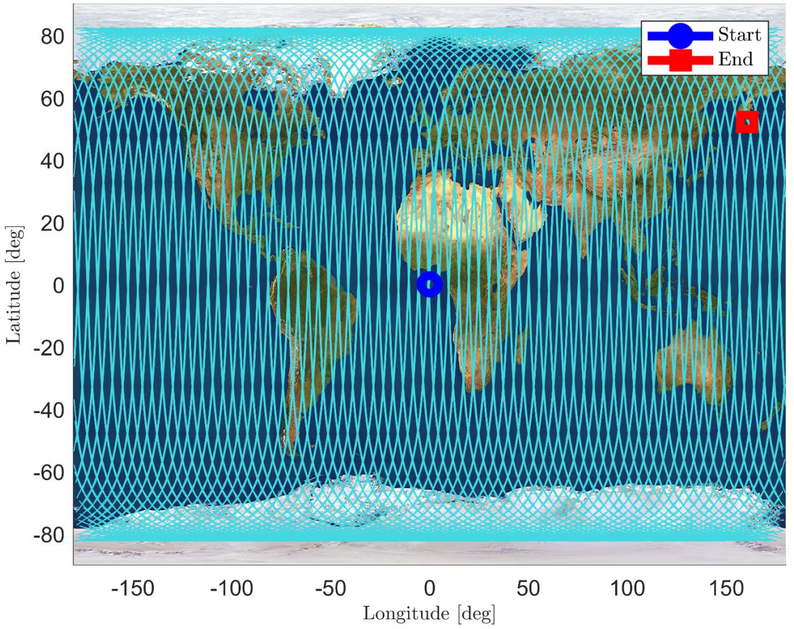
GROUND TRACKS UNPERTURBED NOMINAL ORBIT



1 orbit

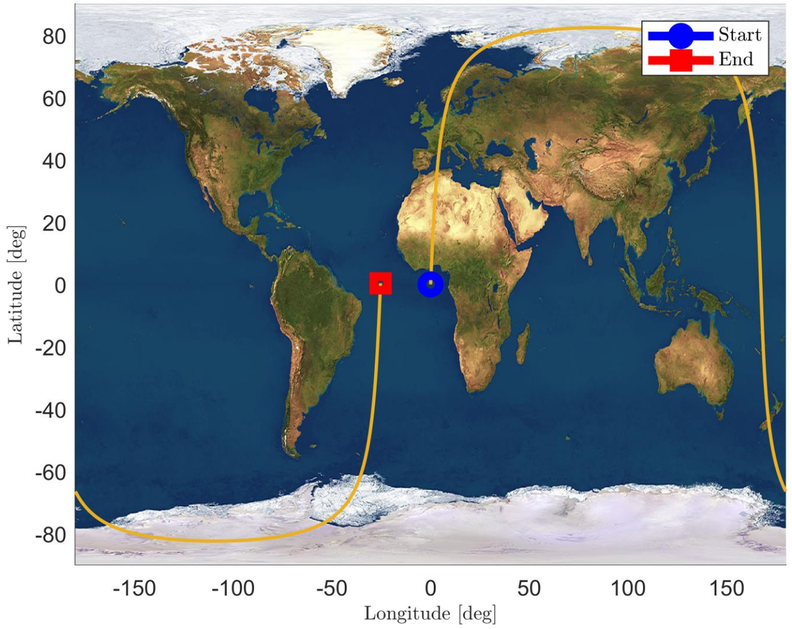


1 day

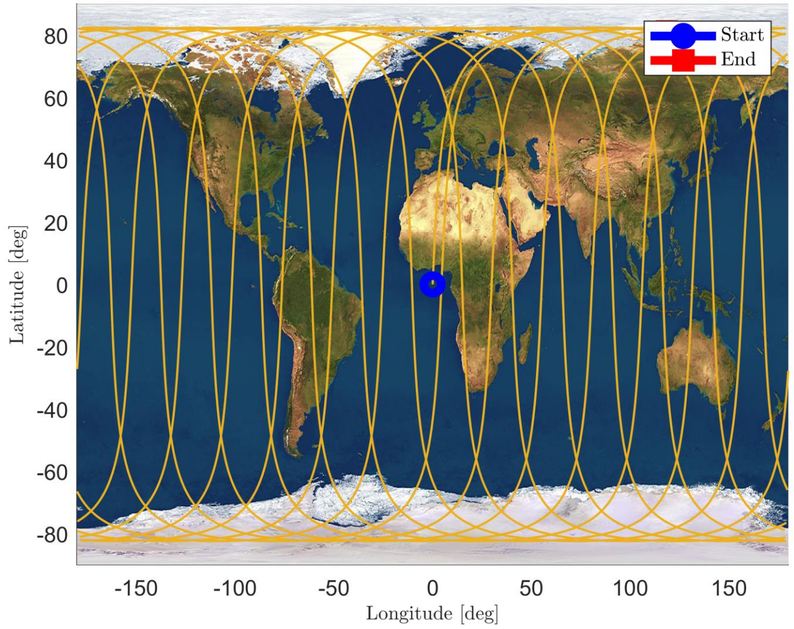


10 days

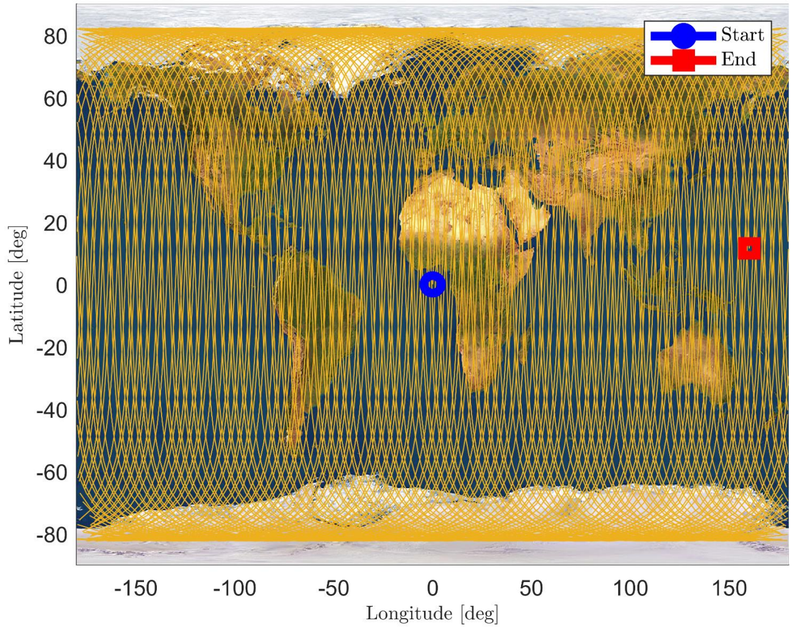
GROUND TRACKS PERTURBED NOMINAL ORBIT



1 orbit



1 day



10 days

REPEATING GROUND TRACK

Regarding the ground track repetition, for the unperturbed motion the only effect to consider is again the rotation of the Earth. The semi-major axis required for the ground track repetition is obtained from the period of a repeating ground track obtained by the following relation.

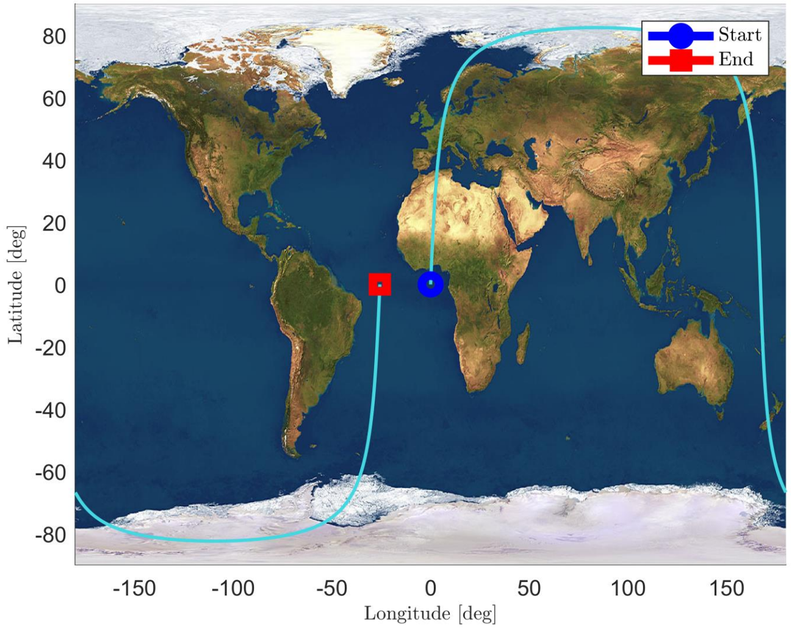
$$T_{rep} = \frac{2\pi}{\omega_E} \cdot \frac{m}{k}$$

where k and m are the number of revolutions of the spacecraft and the Earth respectively after which the repetition must take place.

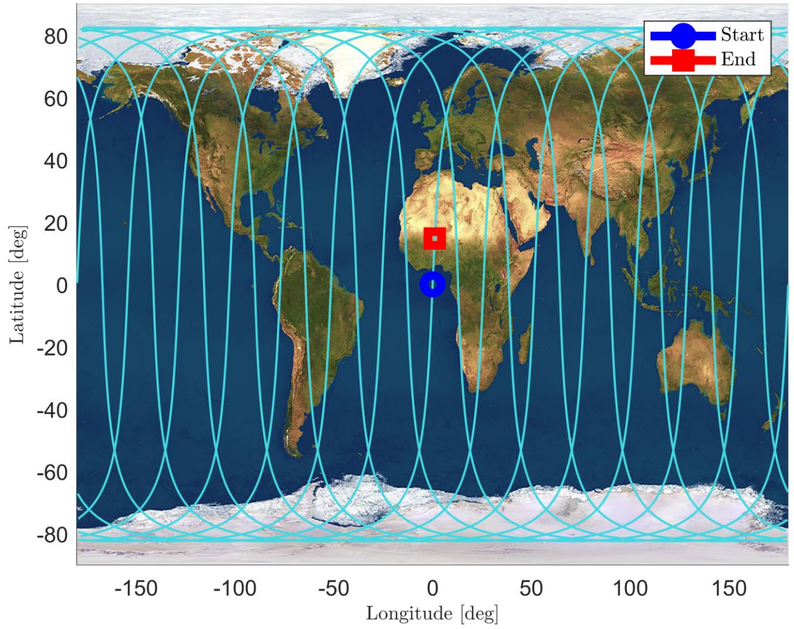
$a_{rep}[km]$	$T_{rep}[h]$
7259.0329	1.7097



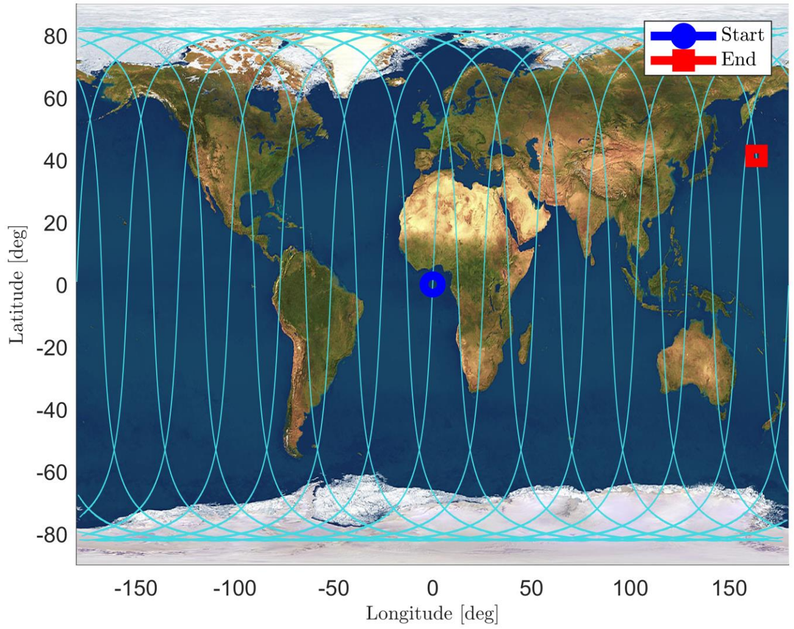
GROUND TRACKS UNPERTURBED REPEATING ORBIT



1 orbit

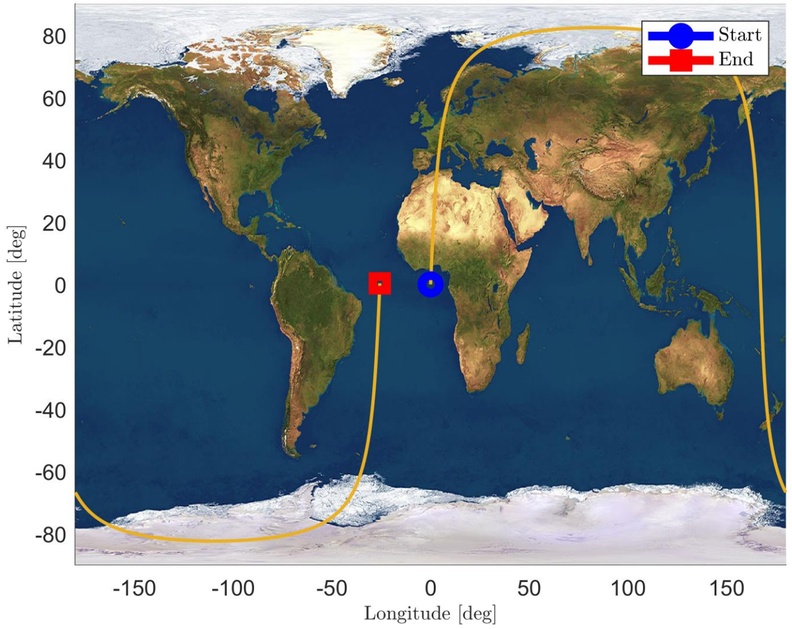


1 day

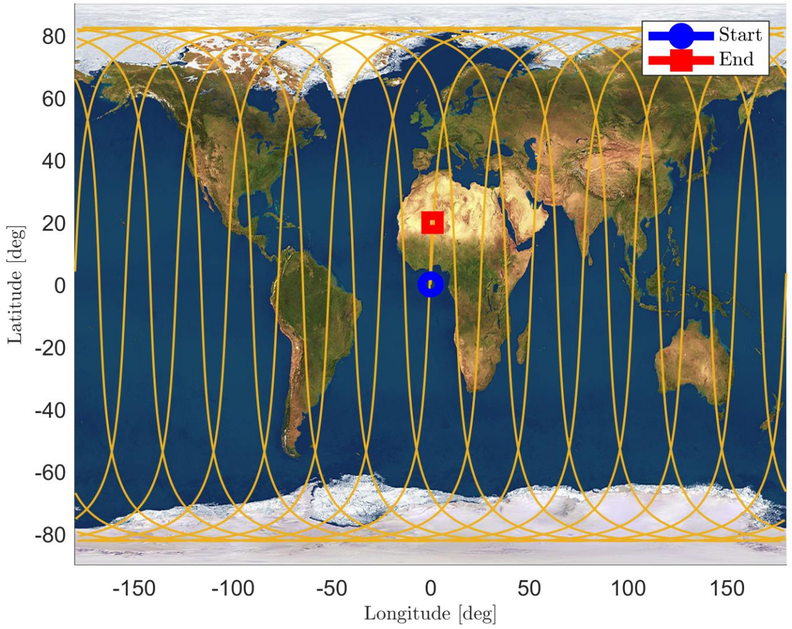


10 days

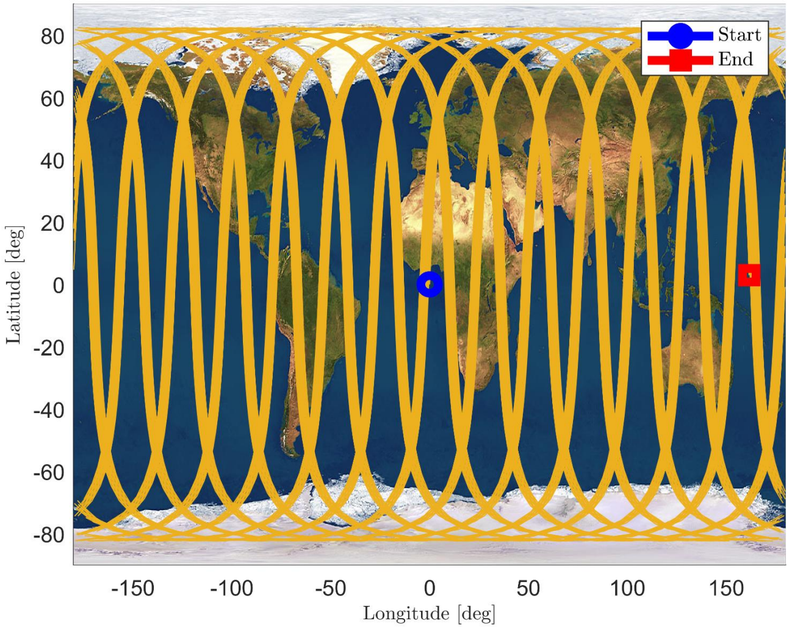
GROUND TRACKS PERTURBED REPEATING ORBIT



1 orbit



1 day



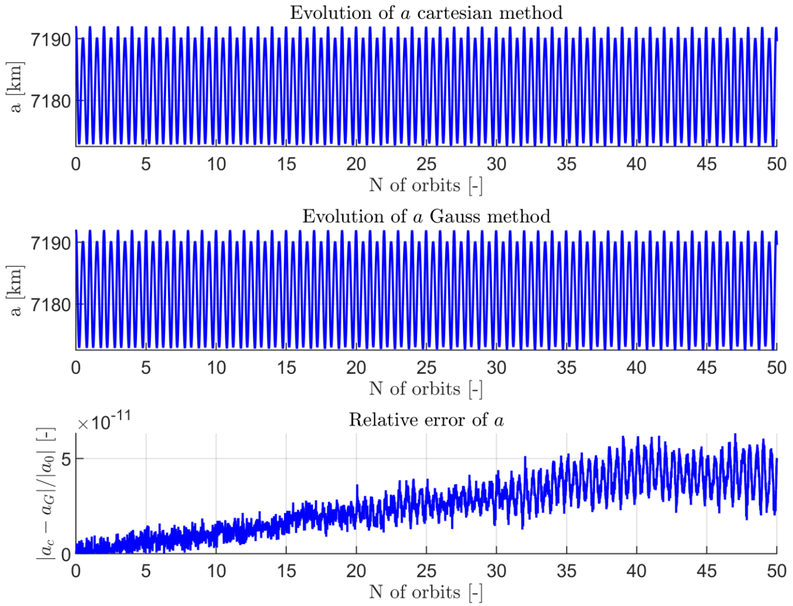
10 days

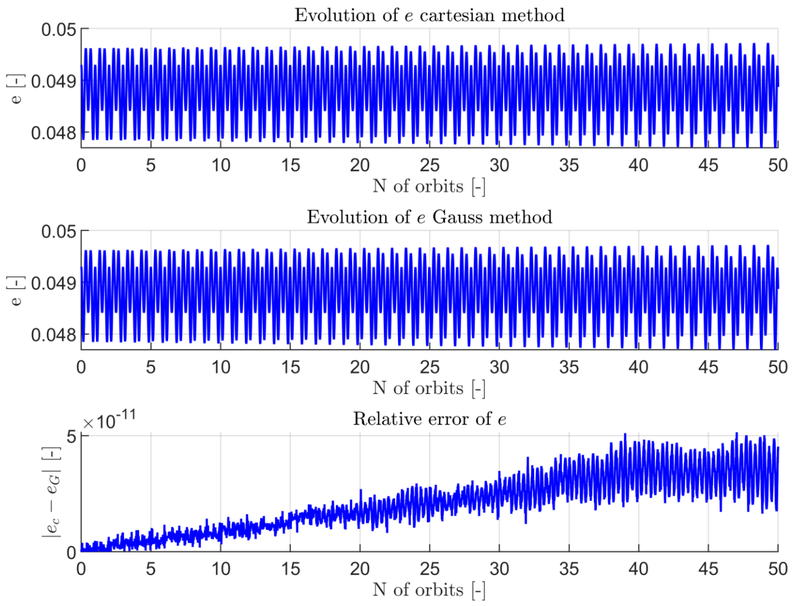
GAUSS PALENTARY EQUATIONS

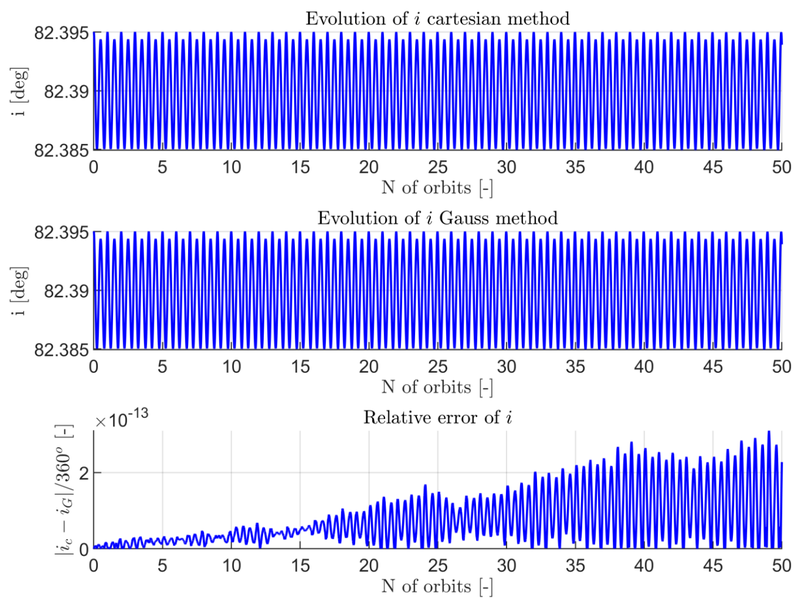
$$\begin{cases} \dfrac{da}{dt} = \dfrac{2a^2 v}{\mu} a_t \\ \dfrac{de}{dt} = \dfrac{1}{v}\left(2(e+\cos\theta)a_t - \dfrac{r}{a} sin\theta a_n\right) \\ \dfrac{di}{dt} = \dfrac{rcos\theta^*}{h} a_h \\ \dfrac{d\Omega}{dt} = \dfrac{rsin\theta^*}{hsini} a_h \\ \dfrac{d\omega}{dt} = \dfrac{1}{ev}\left(2sin\theta a_t + \left(2e + \dfrac{r}{a}cos\theta\right)a_n\right) - \dfrac{rsin\theta^* cosi}{hsini} a_h \\ \dfrac{d\theta}{dt} = \dfrac{h}{r^2} - \dfrac{1}{ev}\left(2sin\theta a_t + \left(2e + \dfrac{r}{a}cos\theta\right)a_n\right) \end{cases}$$

In order to show the propagation of the orbit and evaluating the evolution of its Keplerian elements two methods have been used: the integration of the equation of motion and the integration of the Gauss Planetary Equations[4].

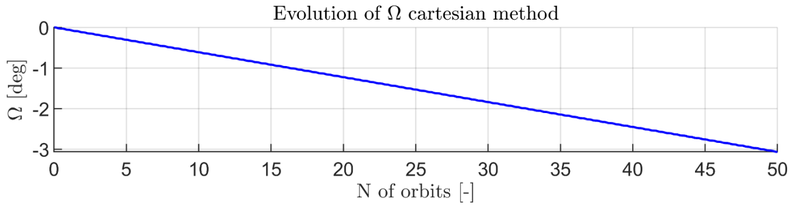
COMPARISON BETWEEN CARTESIAN AND GAUSS PLANETARY EQUATIONS

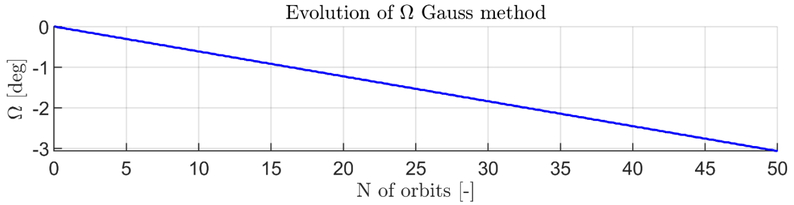


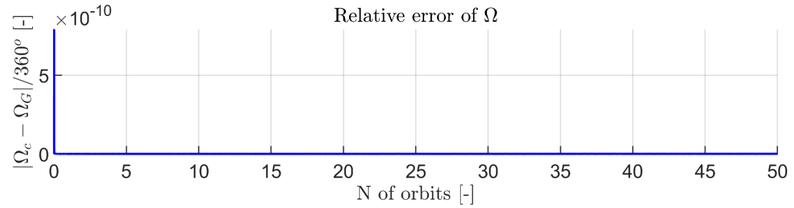


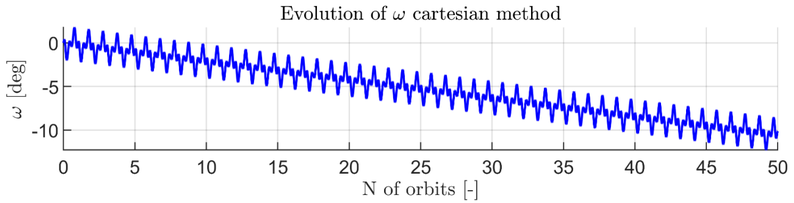


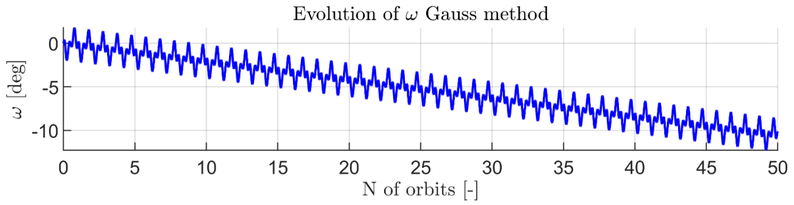
COMPARISON BETWEEN CARTESIAN AND GAUSS PLANETARY EQUATIONS

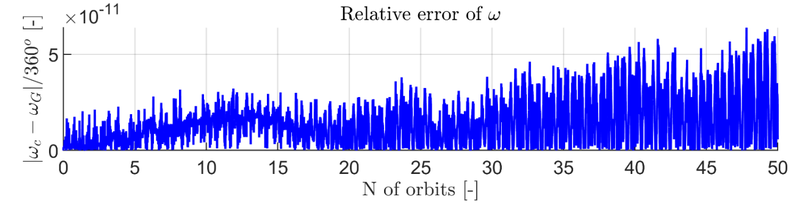


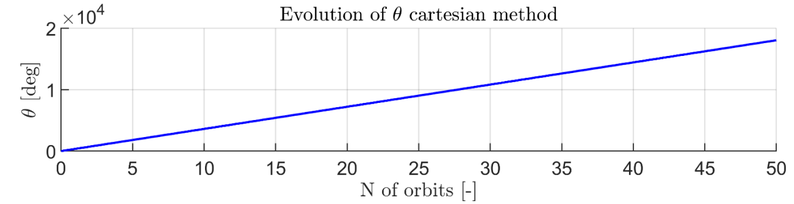


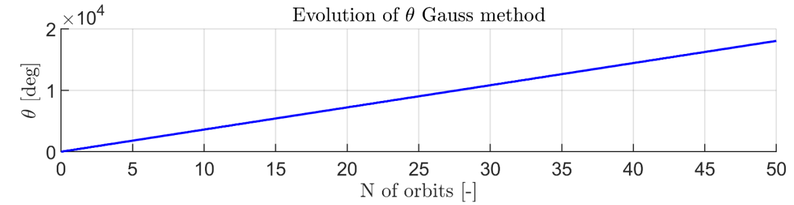


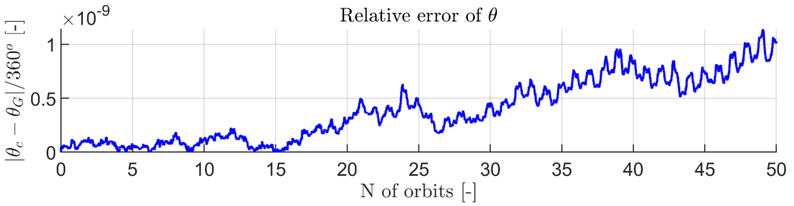












RELATIVE ERRORS

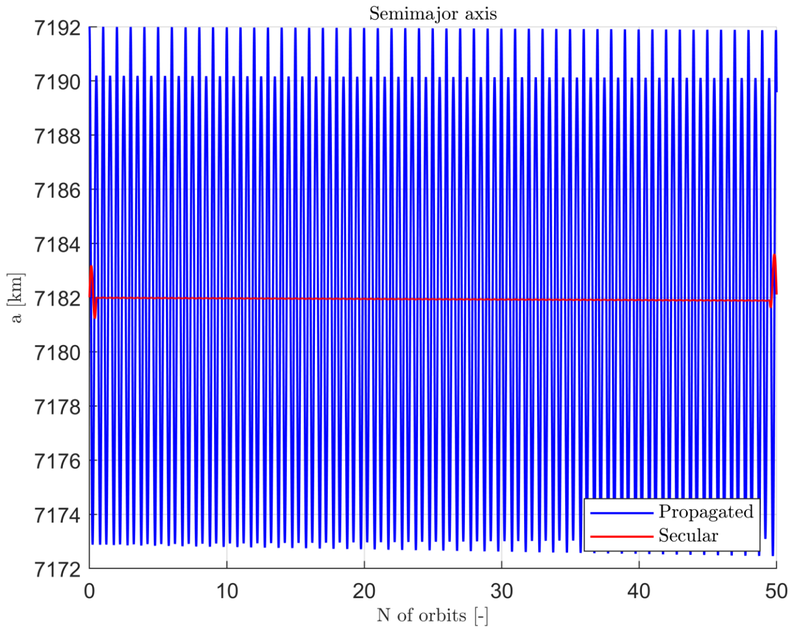
As we just saw, there is virtually no difference in terms of accuracy between the two methods, but as for computational efficiency the Gauss method is faster when a finer discretization of the time interval is used.

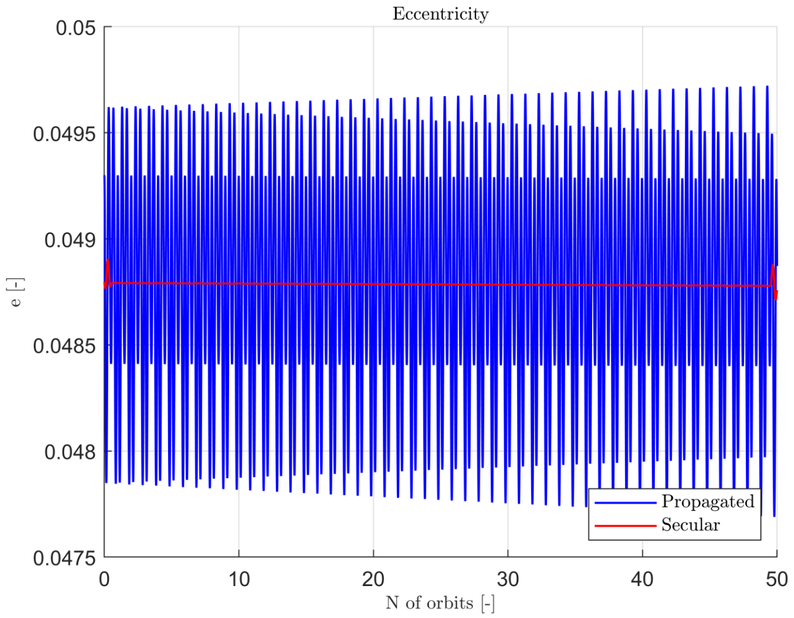
a	e	i	Ω	ω	θ
6.5241e-11	5.4757e-11	3.4855e-13	7.9018e-10	7.6616e-11	9.4066e-09

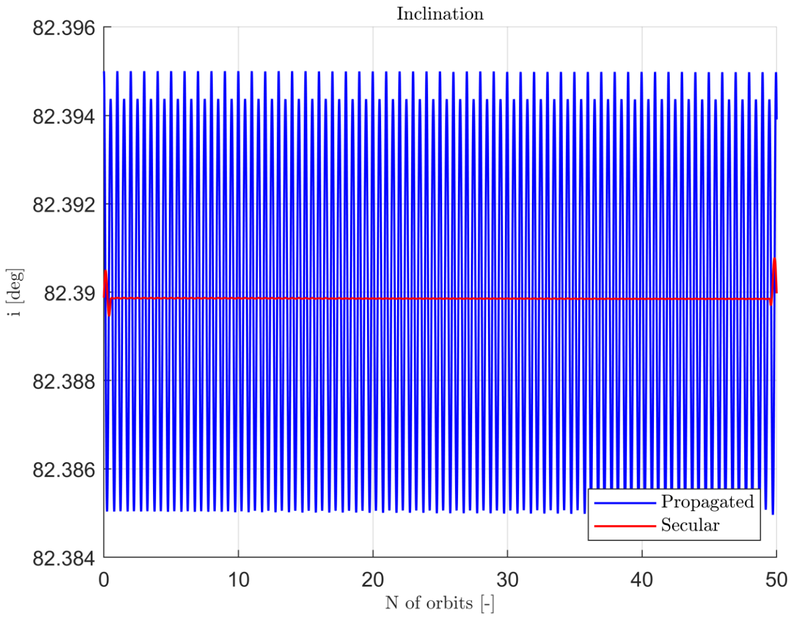
FILTERING

A single oscillatory behavior can be noticed among all the Keplerian elements, which is a short periodic oscillation (with a period of approximately 1 orbit) due to the presence of the J2 and the aerodynamic drag perturbation.
In order to recover only the secular effects, a low-pass filter has been applied to the propagated elements.

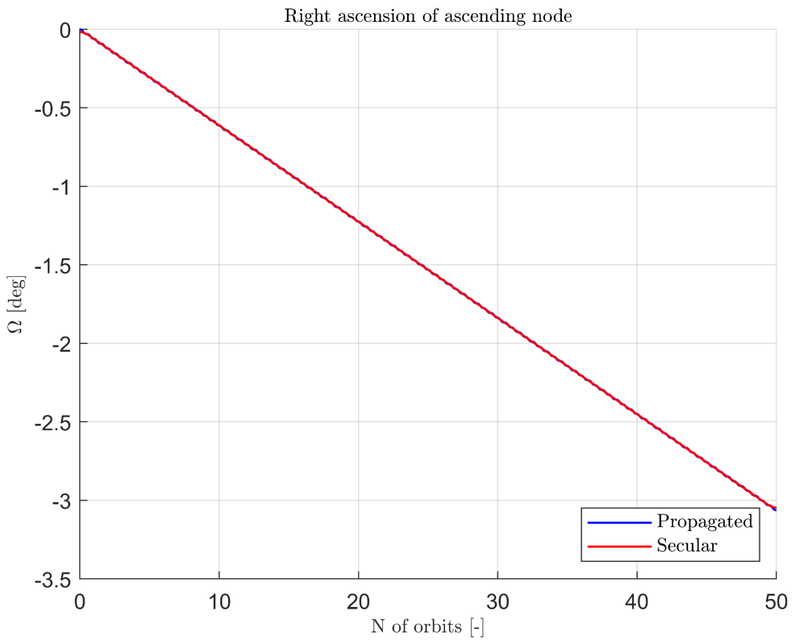
FILTERING OF KEPLERIAN ELEMENTS

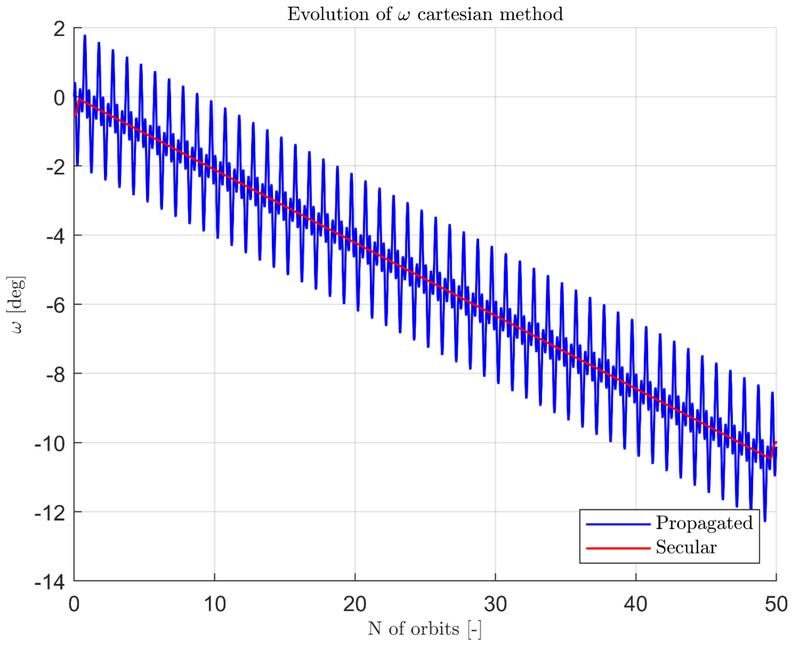


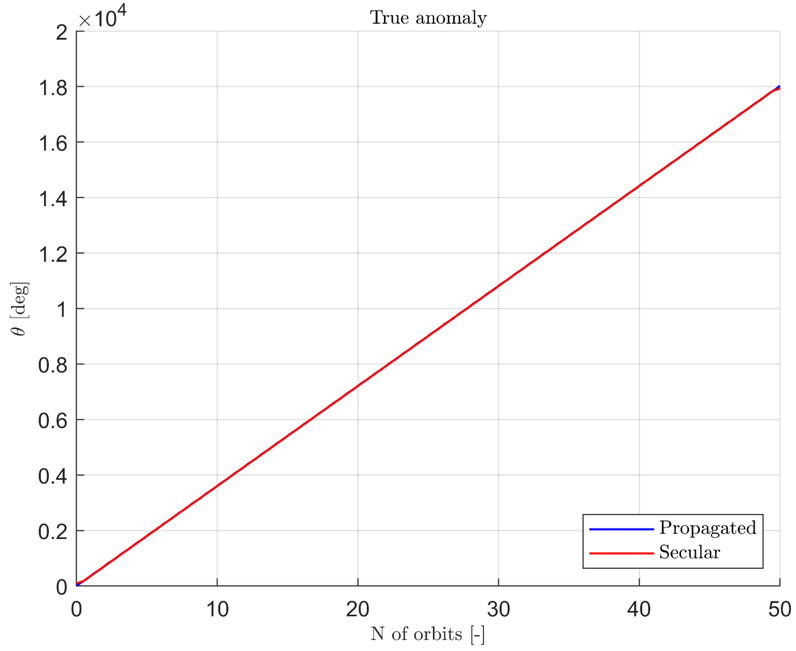




FILTERING OF KEPLERIAN ELEMENTS



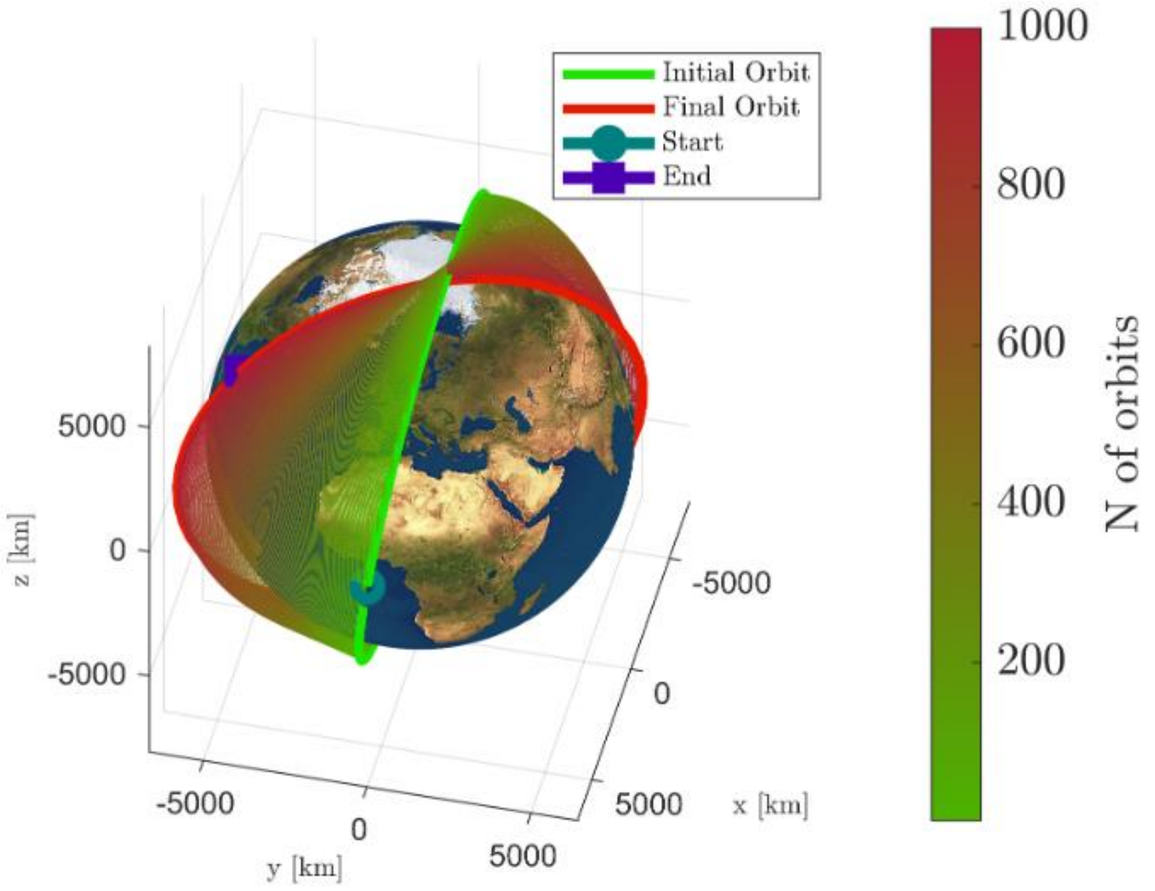




ORBIT EVOLUTION

Evolution of the orbit, propagated with Gauss Planetary Equations, for 1000 orbits which correspond to approximately 70 days.

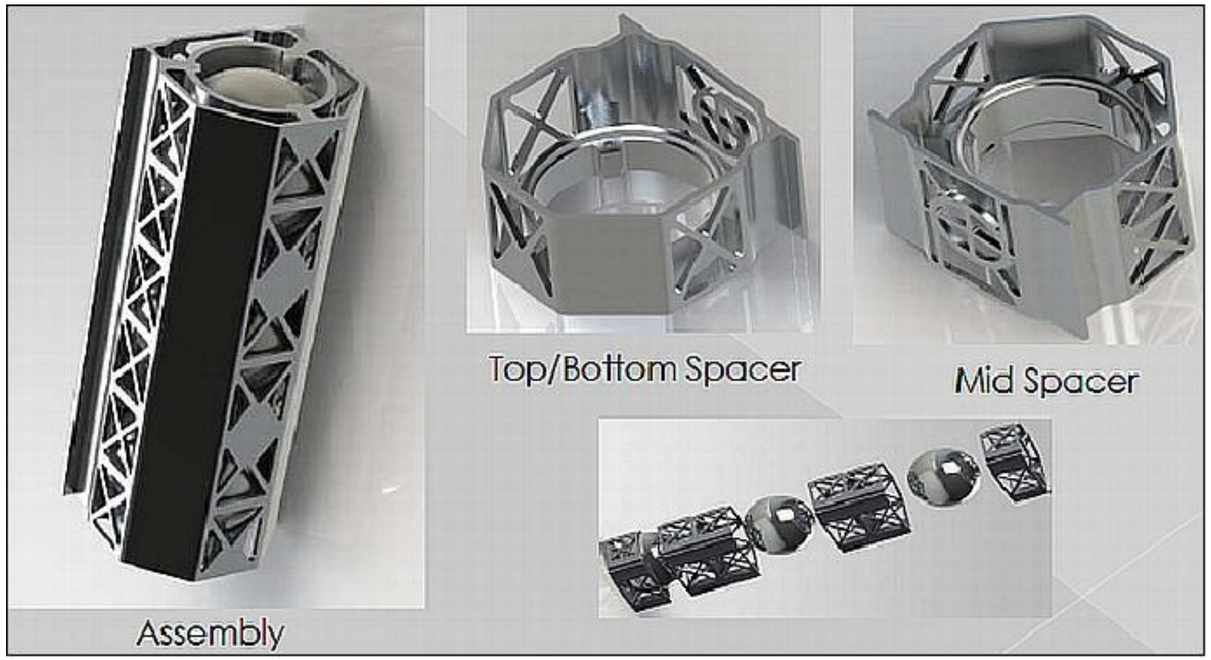
	$a[km]$	$e[-]$	$i[deg]$	$\Omega[deg]$	$\omega[deg]$	$\theta[deg]$
Initial time	7192	0.0493	82.395	0	0	0
Final time	7181.9166	0.0490	82.3905	-61.3212	-211.0398	249.5298



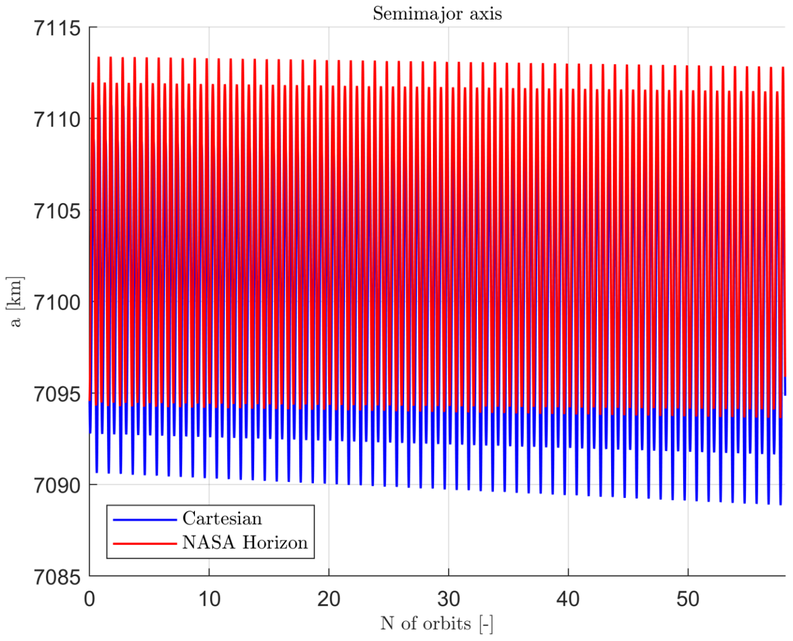
POPACS 2

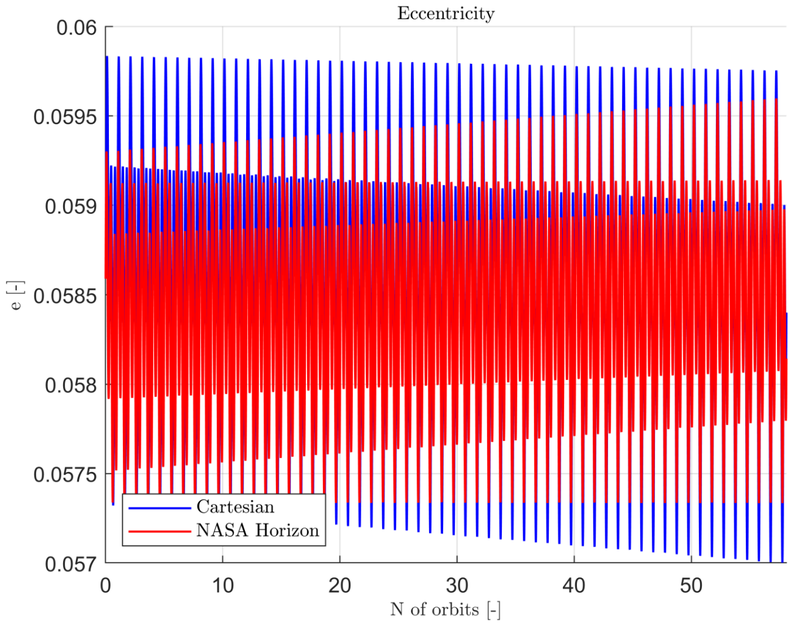
In order to validate the numerical propagation, a comparison has been performed using the data of an accurate model made by NASA Horizons[6]. The POPACS 2[1] has been taken as reference because the orbit is similar to the nominal one.

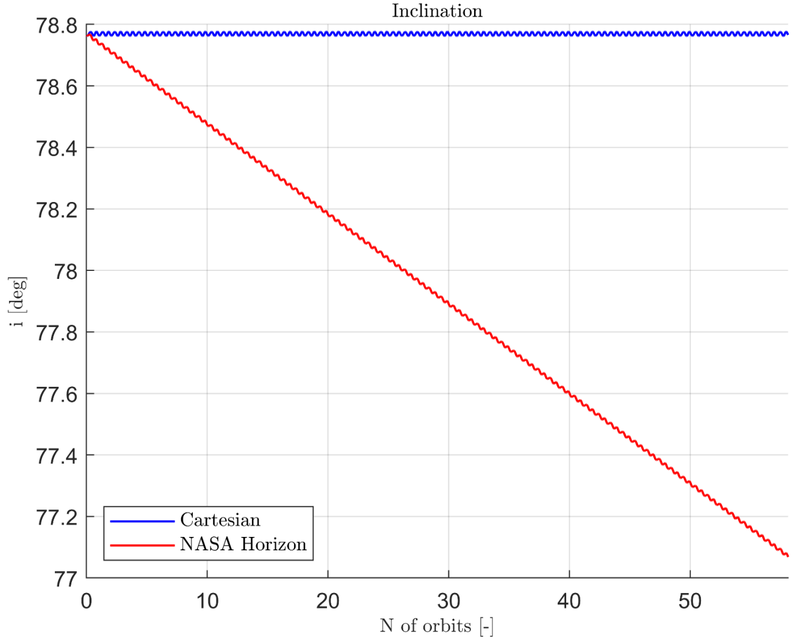
$a[km]$	$e[-]$	$i[deg]$	$\Omega[deg]$	$\omega[deg]$	$\theta_i[deg]$	T [h]
7094.5604	0.0586	78.7644	86.3885	147.4482	100.5591	1.6861



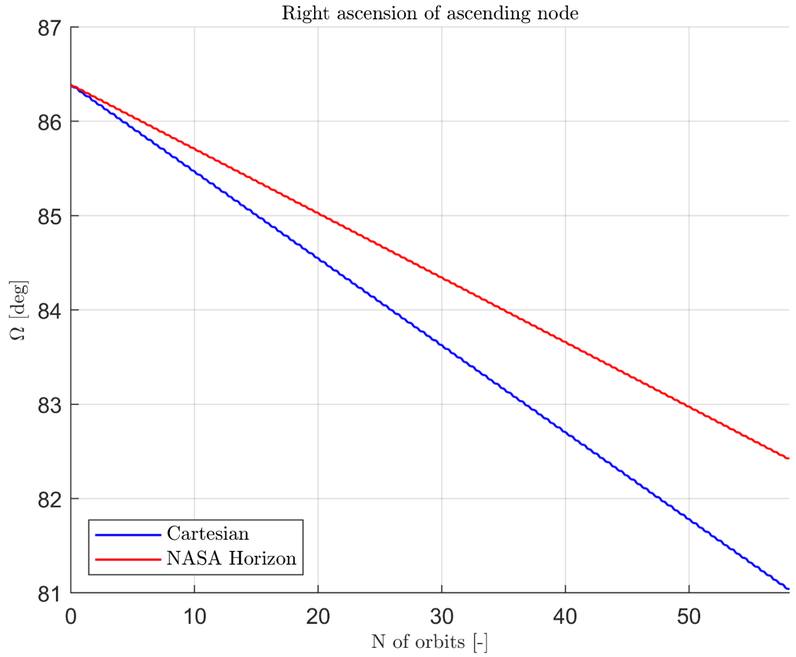
COMPARISON BETWEEN REAL DATA AND CARTESIAN METHOD

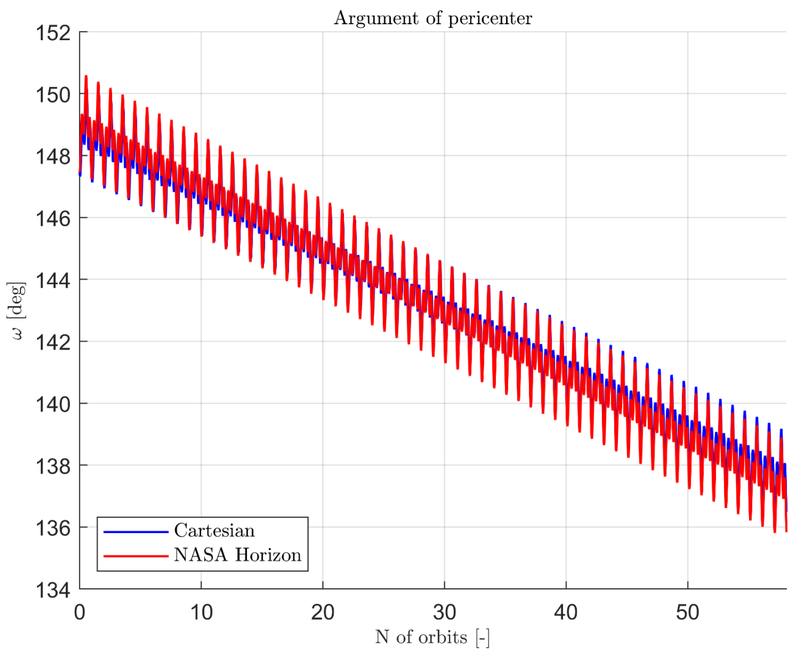


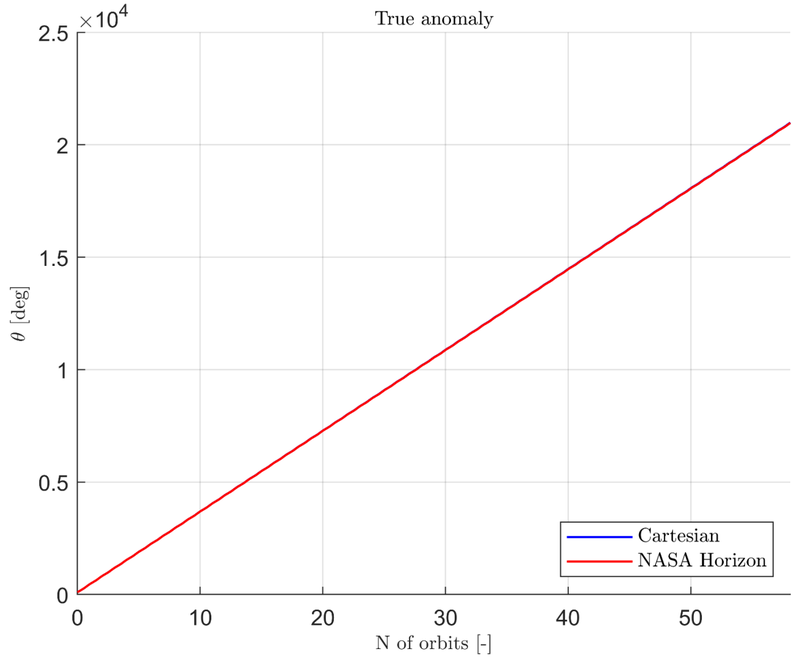




COMPARISON BETWEEN REAL DATA AND CARTESIAN METHOD







ERRORS

The following table shows the errors between cartesian method and NASA Horizons System[6].

a	e	i	Ω	ω	θ
1.5530e-03	1.0506e-03	2.7018e-01	2.2056e-01	1.7354e-01	6.1253e-03

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— [5] Celestrack. https://celestrak.org/NORAD/elements/table.php?GROUP=cubesat& FORMAT=tle.

— [6] Nasa horizon. https://ssd.jpl.nasa.gov/horizons/app.html#/.

THANK FOR YOUR ATTENTION

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